

Weather API Documentation

Base URL: `http://127.0.0.1:8000/api/`
Hosted Base URL: `https://rainl.pythonanywhere.com/api/`

Authentication

All protected endpoints require login. Use `/api/login/` first to get a session cookie.

Method	Endpoint	Description
POST	<code>/api/register/</code>	Create new user
POST	<code>/api/login/</code>	Login
POST	<code>/api/logout/</code>	Logout
GET	<code>/api/check-auth/</code>	Check login status

Register Example

```
POST /api/register/
{"username": "john", "email": "john@test.com", "password": "123"}

201 Created:
{"success": true, "user_id": 1, "username": "john"}

400 Bad Request:
{"success": false, "error": "Username already taken"}
```

Login Example

```
POST /api/login/
{"username": "john", "password": "123"}

200 OK:
{"success": true, "message": "Login successful", "user_id": 1}

401 Unauthorized:
{"success": false, "error": "Invalid username or password"}
```

Weather Data (Public)

Method	Endpoint	Parameters
GET	<code>/api/search/</code>	<code>q=London</code>
GET	<code>/api/weather/</code>	<code>lat=51.5&lon=-0.12</code>
GET	<code>/api/forecast/</code>	<code>lat=51.5&lon=-0.12</code>
GET	<code>/api/pollution/</code>	<code>lat=51.5&lon=-0.12</code>

Search Example

```
GET /api/search/?q=London

200 OK:
{
  "success": true,
  "data": [{"name": "London", "lat": 51.5, "lon": -0.12}]
}
```

Weather Example

```
GET /api/weather/?lat=51.5&lon=-0.12

200 OK:
{
  "success": true,
  "data": {
    "temperature": 11.5,
    "weather": "clouds",
    "humidity": 72
  }
}
```

Favorites (Login Required)

Method	Endpoint	Description
GET	/api/favorites/	Get all favorites
POST	/api/favorites/add/	Add favorite
DELETE	/api/favorites/delete/1/	Delete favorite

Add Favorite Example

```
POST /api/favorites/add/
{"name": "London", "country": "GB", "lat": 51.5, "lon": -0.12}

201 Created:
{"success": true, "favorite_id": 1}

400 Bad Request:
{"success": false, "error": "Already in favorites"}
```

Get Favorites Example

```
GET /api/favorites/

200 OK:
{
  "success": true,
  "favorites": [
    {"id": 1, "name": "London", "lat": 51.5, "lon": -0.12}
  ]
}
```

Weather Alerts (Login Required)

Method	Endpoint	Description
GET	/api/alerts/	Get all alerts
POST	/api/alerts/create/	Create alert
PUT	/api/alerts/toggle/1/	Toggle on/off
DELETE	/api/alerts/delete/1/	Delete alert
GET	/api/alerts/check/1/	Check triggered

Create Alert Example

```
POST /api/alerts/create/
{"city_id": 1, "alert_type": "TEMP_LOW", "threshold": 5.0}

201 Created:
{"success": true, "alert_id": 1}

404 Not Found:
{"success": false, "error": "City not found"}
```

Check Alerts Example

```
GET /api/alerts/check/1/

200 OK:
{
  "success": true,
  "current_temp": 3.2,
  "alerts_triggered": [
    {"id": 1, "type": "Low Temperature", "threshold": 5.0}
  ]
}
```

Analytics (Login Required)

Method	Endpoint	Description
GET	/api/analytics/daily-summary/	Summary over user's favourites

Daily Summary Example

```
GET /api/analytics/daily-summary/

200 OK:
{
  "success": true,
  "summary": {
    "favorite_count": 1,
    "average_temperature": 10.96,
    "coldest": {
      "favorite_id": 3,
      "city": "Paris",
      "country": "FR",
      "lat": 48.85,
      "lon": 2.35,
      "temperature": 10.96,
      "weather": "broken clouds"
    },
    "hottest": {
      "favorite_id": 3,
      "city": "Paris",
      "country": "FR",
      "lat": 48.85,
      "lon": 2.35,
      "temperature": 10.96,
      "weather": "broken clouds"
    }
  },
  "details": [
```

```

    {
      "favorite_id": 3,
      "city": "Paris",
      "country": "FR",
      "lat": 48.85,
      "lon": 2.35,
      "temperature": 10.96,
      "weather": "broken clouds"
    }
  ]
}

```

History (Login Required)

Method	Endpoint	Description
GET	/api/history/weather/	Get recent weather search history for user

Weather History Example

```

GET /api/history/weather/

200 OK:
{
  "success": true,
  "count": 3,
  "history": [
    {
      "city": "Paris",
      "country": "FR",
      "lat": 48.85,
      "lon": 2.35,
      "temperature": 10.9,
      "searched_at": "2026-03-12T19:12:00Z"
    }
  ]
}

```

Status Codes

200	Success
201	Created
400	Bad Request
401	Unauthorized
404	Not Found
405	Method Not Allowed